Identifying NFL Quarterback Archetypes via K-Means Clustering

Shubhan Tamhane | Undergraduate Data Science Student | University of Connecticut | Spring 2026

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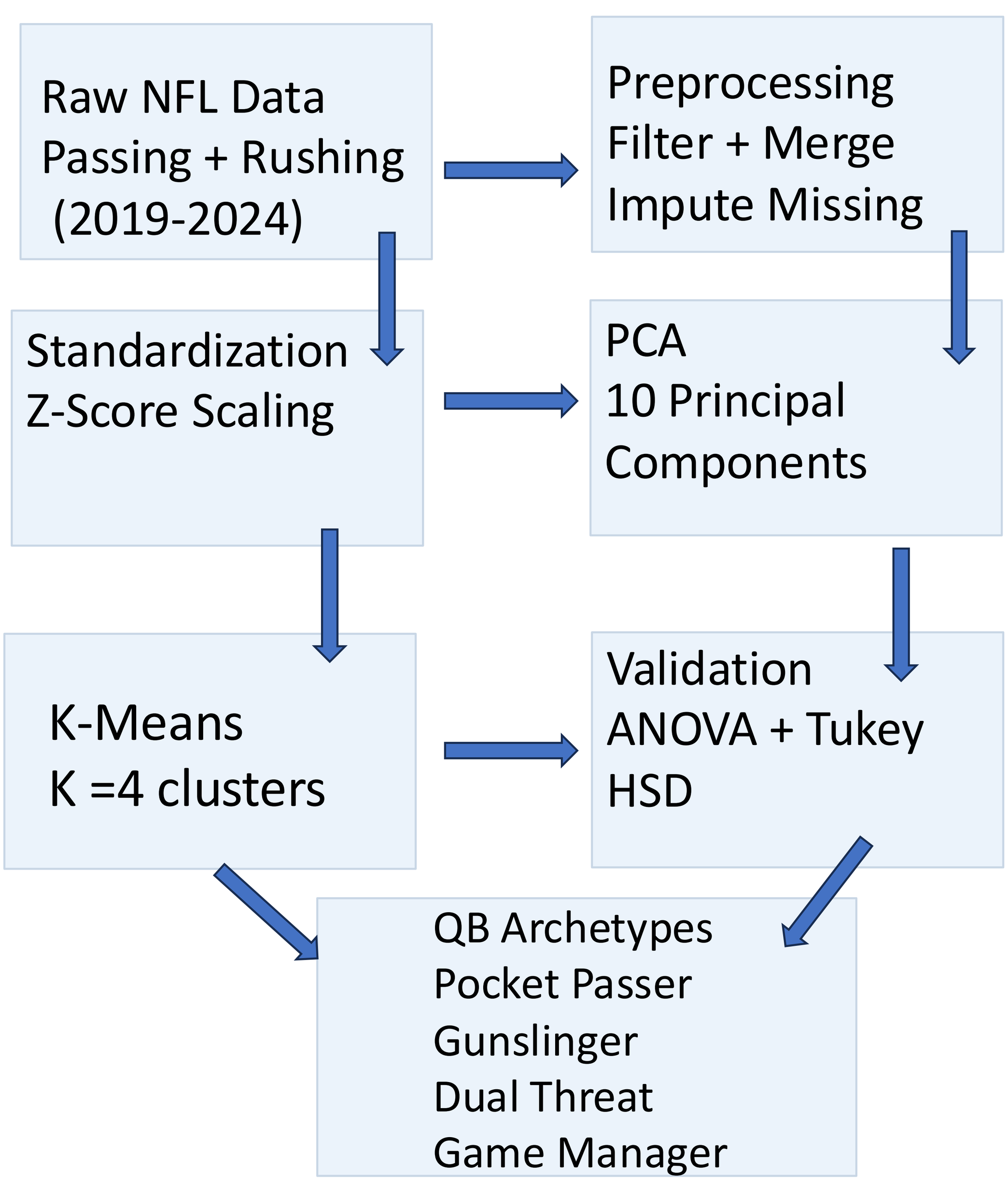
ABSTRACT

Conventional QB metrics like passer rating collapse multidimensional playing styles into a single value. This study asks: Can K-means clustering identify distinct, interpretable archetypes from NFL performance?

DATA

204	2019-24	≥100/20
QB-Seasons	NFL Seasons	Pass/Rush Att.

Source: Pro Football Reference. Merged passing & rushing season stats; mid-season trades handled via totals rows. Missing QBR imputed with mean; longest rush with median.



CONCLUSIONS

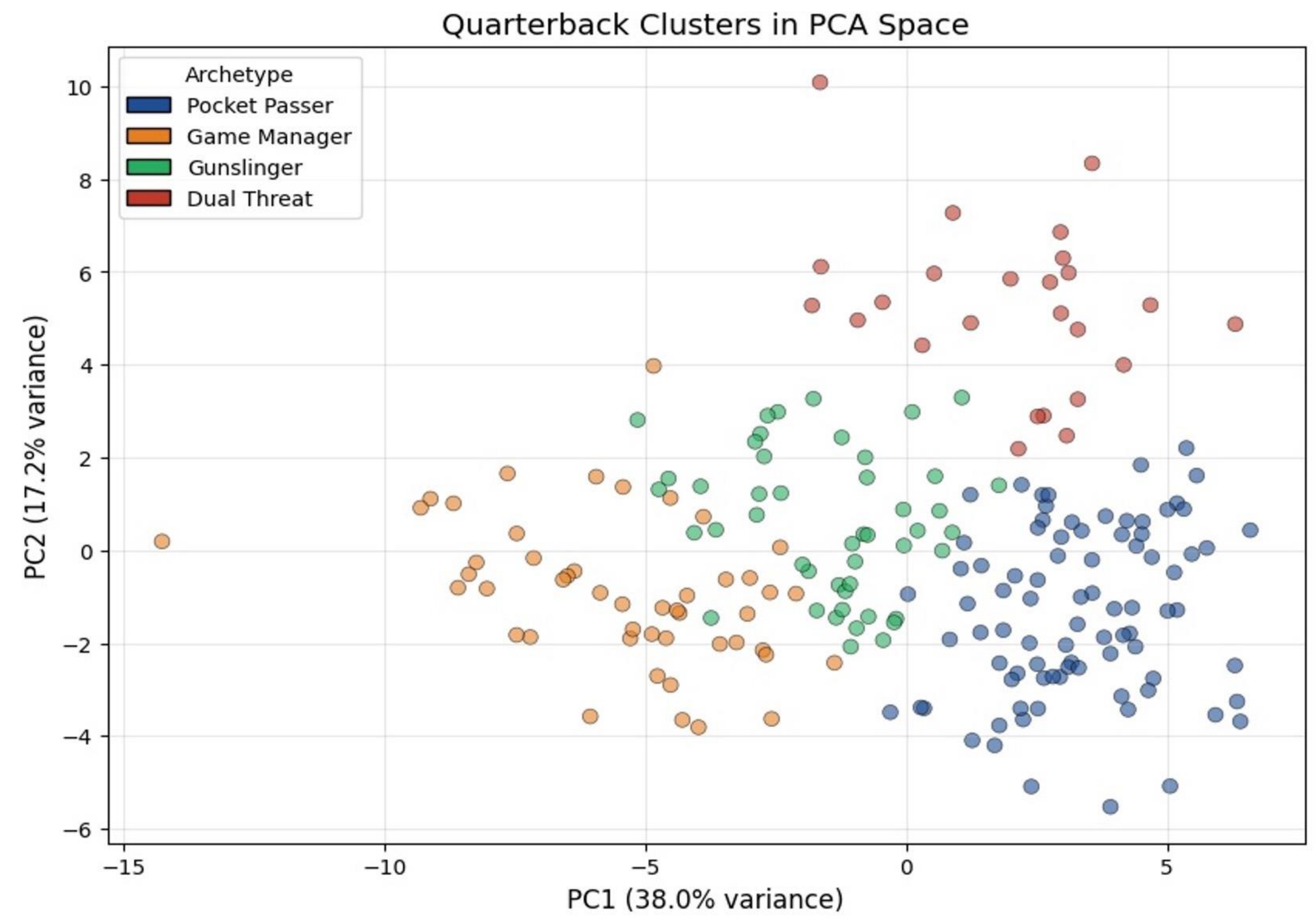
✓ All 10 ANOVA tests significant ($p < 0.001$) — clusters are statistically real.

✓ Pocket Passers and Dual Threats have similar QBR, efficiency achieved via different styles.

⚠ Archetypes exist on a continuum; moderate boundary overlap is expected.

→ Future work: within-career style evolution; Next gen stats

RESULTS — QB CLUSTERS IN PCA SPACE



PC1 (38.0% variance) separates high-volume passers from limited starters. PC2 (17.2% variance) isolates rushing-dominant players. Dual Threat QBs form a visually distinct cluster at high PC2 values.

CLUSTER MEANS (KEY METRICS)

Archetype	Pass Yds	TD (pass)	INT	QBR	Rush Yds	Rush TD
Pocket Passer	4,102	28.3	10.5	**61.8**	172	1.5
Game Manager	1,475	8.1	5.7	43.4	139	0.9
Gunslinger	2,903	15.1	10.8	46.4	237	1.2
Dual Threat	3,430	23.8	9.8	**63.0**	**720**	**5.8**

EXAMPLE PLAYERS PER ARCHETYPE

Pocket Passer: Patrick Mahomes (KC), Dak Prescott (DAL), Joe Burrow (CIN), Tom Brady (NE)

Game Manager: Mac Jones (NE), Devlin Hodges (PIT) , Early Jalen Hurts (2020, PHI)

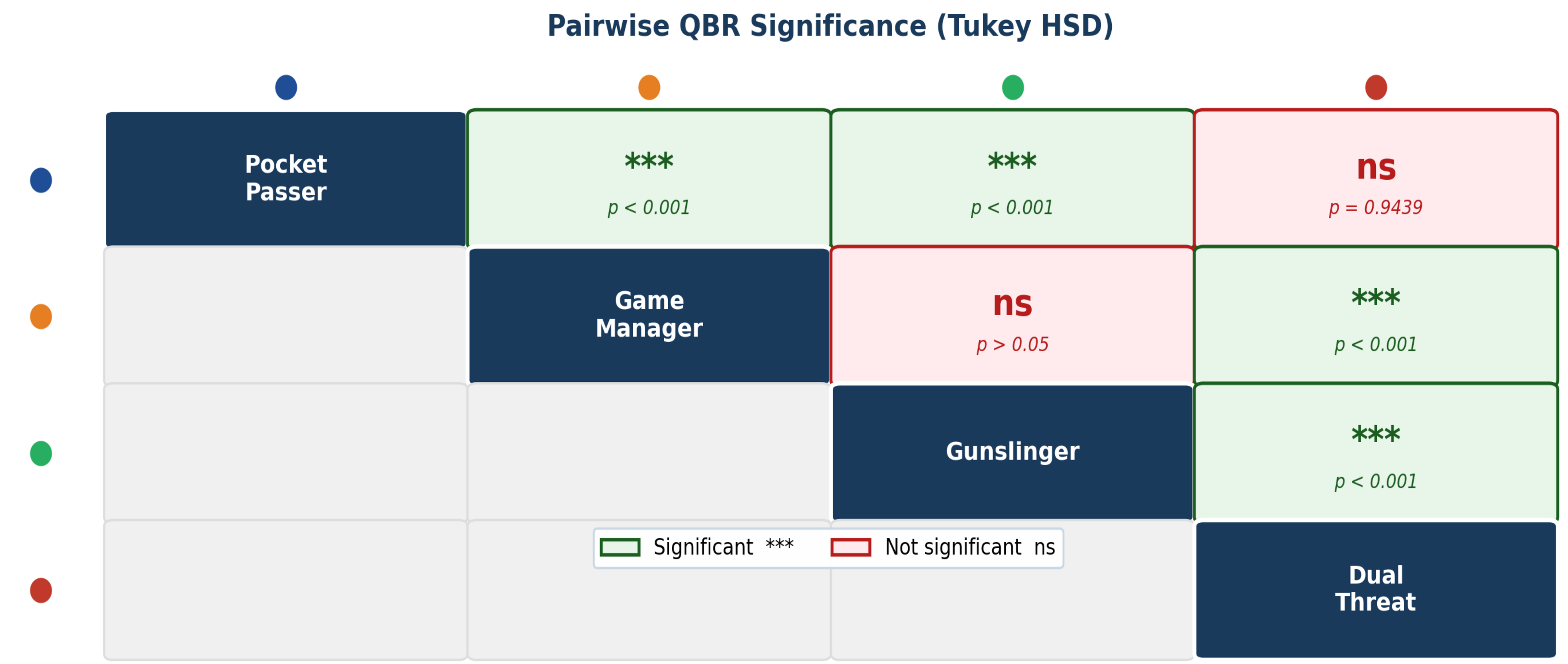
Gunslinger: Justin Herbert (LAC), Taylor Heinicke (WAS) , Sam Howell (WAS) , Josh Johnson (DEN)

Dual Threat: Lamar Jackson (BAL) , Kyler Murray (ARI) , Josh Allen (BUF) , Jalen Hurts (PHI)

STATISTICAL VALIDATION

10 / 10 metrics statistically significant by one-way ANOVA (all $p < 0.001$)

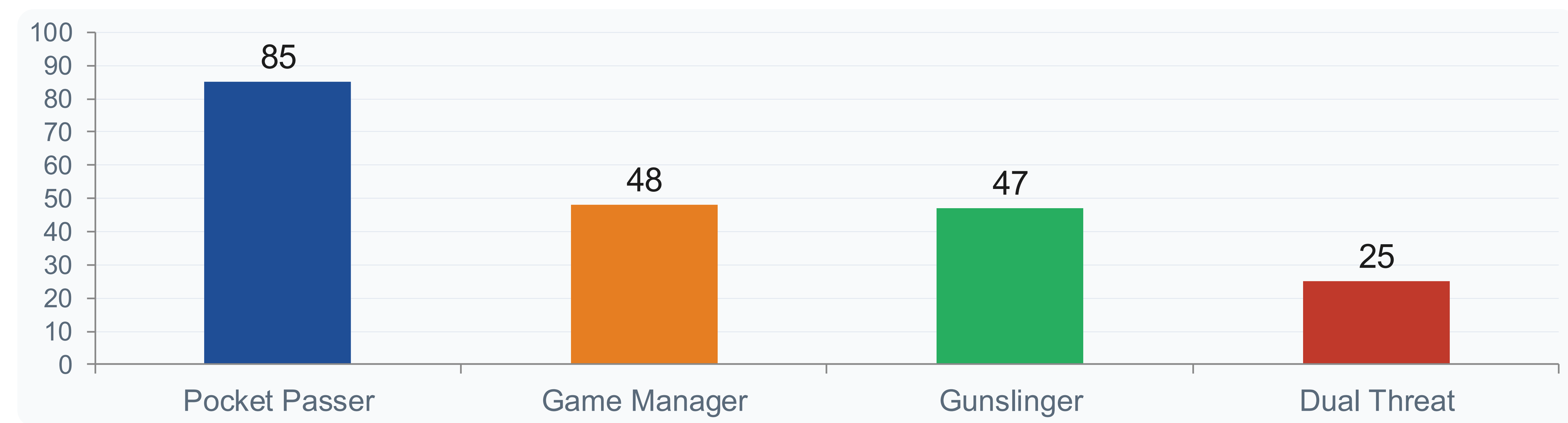
Silhouette score at k=4: 0.208 | Elbow + silhouette both confirmed k=4 | Tukey HSD run on all 10 metrics



💡 Key Finding

Pocket Passers and Dual Threats show no significant difference in QBR ($p = 0.9439$), despite radically different playing styles, elite efficiency is achievable through passing volume or rushing dominance equally.

ARCHETYPE DISTRIBUTION



CODE AND REPRODUCIBILITY

GitHub Repository

github.com/ShubhanTamhane1/
nfl-qb-archetypes

Scan for source code, notebook, and data pipeline.

References: Celebi et al. (2013) Expert Syst. Appl. 40:200–210 • Lutz (2012) J. Quant. Anal. Sports 8 • Quealy & Carter (2013) NYT • Data: Pro Football Reference (pfr.com)